Decoding magnetic-field and spaceflight stress signatures onto the dry/germinating Arabidopsis seed

A cell-type-resolved bridge decoder — Phase 4 synthesis

Date: 2026-06-26 · **Project:** NMF meta-analysis + bridge seed decoder (standalone) **Status:** decoder + bridge + NMF localization functional end-to-end; genome-wide NMF pending.

Evidence tiers used throughout: [D] Direct = genome-wide differential expression we projected · **[A] Atlas** = single-cell seed-atlas panels / pseudobulk · **[L] Literature** = published cell-type identities, NMF gene directions, magnetobiology · **[H] Hypothesis** = predicted concordance + falsification test. All decoder/bridge outputs are **concordance / signature transfer, not proven causation.**

1. Executive summary

1. We built a **122-panel seed reference** [A] spanning the developing seed (Gehring snRNA atlas, GSE295007: embryo/endosperm/seed-coat, 5 tissues → 13 cell types → 86 states) and the **dry→germinating seed** (germination atlas, GSE182331/E-MTAB-12532: 15 named cell types across 12/24/48 h). Panels validated against canonical markers (OLE/CRA/2S → embryo; BAN → endothelium).
2. A **projection decoder** [D×A] maps genome-wide spaceflight/radiation signatures (NASA OSDR GLDS-120 root µg, GLDS-612 leaf µg, GLDS-603 GCR) onto these seed programs via GSEA-prerank.
3. **Stressed adult/vegetative tissue transcriptionally resembles the early-germination (12 h) state** — the most consistently induced program across µg and GCR-80 [D×A].
4. A **shared "germinating-seed score space" bridge** places both adult-stress and late-seed-dev inputs relative to the dry/germinating reference. The **dev→germination bridge is lineage-limited**: only the embryo lineage transfers; maternal seed coat and endosperm are terminal [A×L]. 4b. **Positive control:** restricting the dev bridge to the embryo lineage recovers **textbook developmental lineages** (protoderm→epidermis, hypophysis→radicle apical meristem, vascular primordium→provasculature) from transcriptomes alone — validating the decoder/bridge machinery [A].
5. **Null magnetic field (NMF) genes localize sharply to the radicle apical meristem** ($z = +7.96$) [L×A]. Together with the light-gated µg radicle signal [D] and the high-field SMF root-meristem/auxin phenotype [L], **multiple magnetic/space stressors converge on the radicle growth-point program** — the central falsifiable hypothesis of this work.

2. Background & objective

Successful germination depends on transcriptional programs laid down across seed development, held through desiccation/dormancy, and re-activated on imbibition. Space-relevant stressors — microgravity (µg), galactic cosmic radiation (GCR), and **null/near-null magnetic field (NMF)** — perturb the adult plant transcriptome, but their consequences for the **dry and germinating seed** are largely uncharacterized, because those experiments measure later-stage tissues, not seeds.

Objective. (i) A comprehensive view of Arabidopsis NMF transcriptional data; and (ii) a **bridge decoder** that takes RNA-seq from later-developmental-stage tissues (adult/vegetative *and* late-seed-development) and projects it onto a dry/germinating-seed reference, predicting which seed cell-type/state programs are concordantly perturbed.

This extends the prior Biomni integrated analysis, which (a) used only the Maffei 194-gene NMF panel and (b) had no seed-resolved data (embryo/endosperm/seed-coat were literature-only). Both gaps are addressed here.

3. Data & methods

3.1 Seed reference atlases [A]

- **Developmental:** Gehring lab, *Nat Plants* 2026 (GSE295007) — snRNA-seq 3/5/7 DAP; 23,374 genes × 54,210 nuclei; annotation hierarchy level_1 (5 tissues) → level_2 (13 cell types) → level_3 (86 states)
- 43 GO module scores.
- **Dry→germinating:** Liew/Lewsey *Nat Plants* 2024 (E-MTAB-12532); expression matrix via GEO mirror **GSE182331** (13,501 genes × 12,798 cells; 15 clusters; 12/24/48 h). Cluster→cell-type identities from the paper's open-access text [L] (clusters 9 & 14 in-situ validated; 3/6/15 confirmed by our markers).

3.2 Panel library [A]

Wilcoxon `rank_genes_groups` (top-50/group, ≥20 cells) → **122 panels / 6,100 marker rows**: Gehring L1 tissue (5), L2 cell type (13), L3 state (86); germination cluster (15), germination time (3).

3.3 Stressor inputs (decoder queries) [D]

Genome-wide DGE from NASA OSDR (TAIR/AGI IDs): **OSD-120/GLDS-120** root µg (dark/light), **OSD-678/GLDS-612** leaf µg (dark/light), **OSD-658/GLDS-603** GCR (40 & 80 cGy). Spaceflight effect = –(ground-vs-flight); radiation effect = irradiated-vs-non.

3.4 Decoder (projection backbone)

For each stressor contrast, rank genome-wide log2FC and run **GSEA-prerank** against all 122 panels (200 permutations). NES > 0 = seed program concordantly **induced**; NES < 0 = **suppressed**; FDR < 0.25 = significant.

3.5 Bridge / shared latent (Layer 2)

A **"germinating-seed score space"** with the 15 named germinating-seed cell types as axes. Adult stress is placed by its decoder NES across those axes; **late-seed-dev** (Gehring tissue×timepoint pseudobulk) is placed by **ssGSEA** against the same 15 panels. Both then sit relative to the dry/germinating reference.

3.6 NMF localization [L×A]

The Maffei NMF panel (~194–230 directional genes) overlaps the 50-gene marker panels by ≤**7 genes (median 0)** → GSEA-prerank infeasible. Instead, NMF-up (198) / NMF-down (18) gene sets were **localized** onto germinating-seed cell-type expression specificity (z vs 1,000 random gene sets). Directions from Maffei cluster-profile (shoot, late timepoints) + polyphenol/H■O■ per-gene log2 ratios.

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4. Results

4.1 Validated panel library [A]

Embryo panels top out on **OLE1/2/3, CRA1, 2S albumin**; inner-integument **ii1 = BAN (AT1G61720)** — the canonical endothelium marker the atlas itself names. Pipeline is biologically sound.

4.2 Stress → seed decoder [D×A]

(Figures: `results/figures/decoder_L1_state_heatmap.png`, `decoder_germination_named_heatmap.png`.)

Tissue-level NES (Gehring L1) and germination-state NES:

program	µg root dark	µg root light	µg leaf dark	µg leaf light	GCR 40	GCR 80
Embryo	+1.07	–1.42	–1.08	+1.38	–1.38	**+2.18**
Endosperm	+0.71	–1.61	–0.94	–1.01	–0.82	+1.28
Seed coat	+1.24	–1.53	–0.98	–1.36	–1.05	–1.47
germ 12 h	+1.65	+1.94	–1.31	+1.92	–1.65	**+2.08**

germ 24 h	+1.17	+1.64	+1.01	+1.59	+1.41	+1.88
germ 48 h	+1.36	–1.44	–0.86	+0.98	–0.66	+0.93

Key patterns: - **Early-germination state (12–24 h) is the dominant induced program** across most stressors — adult stress signatures resemble the early-imbibition transcriptome. - **GCR-80 cGy strongly induces embryo programs** (Embryo +2.18; at L3, EMB vascular primordium +2.45, EMB hypophysis +2.34) — a provascular/founder-cell shift under high radiation. - **Proliferative seed states suppressed** (EMB/PEN g2/m-phase) under µg-root and GCR. - **Strong light-dependence** — µg-root flips sign dark→light for Embryo/Endosperm/Seed coat, echoing the prior Biomni light×treatment interaction.

4.3 Bridge / shared latent [D/A × A]

(Figures: `results/figures/bridge_heatmap.png`, `bridge_embedding.png`.)

Adult-stress → germinating-seed cell type (strong, z 1.4–2.1): µg-leaf-dark → hypocotyl cortex (mid) (+2.05); µg-root-dark → cortex/endodermis (+1.96); GCR-40 → epidermis (+2.00). (GCR-80 and the *light* µg conditions land on "unassigned" clusters 8/11 — soft hits, since those clusters lack a defined identity.)

Late-seed-dev → germinating-seed cell type (weak, z 0.37–0.68) — the weakness is the result: developmental tissues map only weakly/non-specifically, concentrating on root-pole identities (columella, protoxylem, radicle epidermis) for embryo/endosperm 3–7 DAP. **Interpretation:** only the **embryo lineage** persists development → dormancy → germination; the **maternal seed coat and endosperm are terminal**, with no germinating-embryo counterpart. The bridge is real but lineage-limited.

Bridge v3 (canonical — within-source scaling) — extended to all 15 contrasts (microgravity + radiation) + the dev trajectory (27 inputs), with each germ axis z-scored *within source* to remove the score-type artifact. **Diagnostic:** PC1↔source |corr| dropped **0.56** → **0.00**, so the embedding now reflects biology, not data modality (`results/figures/bridge_{heatmap,embedding}_v3.png`, `results/bridge_results_v3.md`). Now interpretable: **Embryo 3 DAP** → **hypocotyl cortex (early), Embryo 5 DAP** → **hypocotyl cortex (mid) —** a developmental progression along the embryo lineage that carries into germination; maternal seed coat/endosperm map to mixed/terminal identities. Radiation favors root-pole / provascular identities (columella, protophloem). **Honesty note:** the perturbation→cell-type *argmax* is scaling-sensitive (shifts v1/v2/v3) → bridge stress assignments are suggestive, not robust; the robust magnetic→seed signal remains the **NMF-up** → **radicle apical meristem localization (z +7.96)**, which is direct expression specificity and scaling-independent.

4.3b Embryo-lineage bridge — developmental validation (positive control) [A]

(Figure: `results/figures/embryo_lineage_heatmap.png`.)

Restricting the dev side to **embryo cells only** (the lineage that persists into germination) and mapping developing-embryo *states* (Gehring level_3) onto germinating-seed cell types **recovers textbook developmental lineages from transcriptomes alone:** - **EMB protoderm** → **epidermis** (z +1.6); **EMB hypophysis** → **radicle apical meristem** (z +2.14); **EMB vascular primordium** → **provasculature/protoxylem** (z +1.88); **EMB inner cotyledon** → **cotyledon mesophyll**; **EMB cortical initials** → **hypocotyl cortex**. Cell-cycle/initial states map to "unassigned".

This is a **positive control**: the decoder+bridge independently re-derives established embryo→seedling lineage relationships, evidencing the machinery is sound. It also **reinforces the radicle convergence** (§4.5): the hypophysis→radicle-apical-meristem edge means the structure NMF/µg/SMF converge on is *founded by the embryonic hypophysis* — a developmental-lineage root for the central hypothesis. (`results/embryo_lineage_results.md`)

4.4 NMF localization [L×A]

(Figure: `results/figures/nmf_localization_heatmap.png`.)

NMF-induced genes concentrate in germinating-seed cell types:

germinating-seed cell type	NMF-up localization z

radicle apical meristem (cl14)	**+7.96**
unassigned (cl8)	+3.23
cotyledon mesophyll (cl13)	+2.85
cortex/endodermis (cl2)	+1.99

NMF-down (n=18, underpowered) → cotyledon mesophyll, cortex/endodermis.

4.6 Radiation / ROS perturbations (expanded panel) [D×A]

(Figure: `results/figures/decoder_combined_perturbation_heatmap.png`.)

The model was extended with WT irradiated-vs-control contrasts from 5 OSDR γ-radiation RNA-seq studies (`radiation_and_ros_cleaned.csv`) → **combined 15-contrast environmental-perturbation model** (microgravity ×4, GCR ×2, low-dose γ ×2, acute 100 Gy γ ×7). Each study also carries an OSD_ID for deeper OSDR data; `sog1-1`/`myb3r1` DNA-damage-TF mutant arms are available for mechanistic follow-up. - **Acute 100 Gy γ (90 min) induces embryo + early-germination (12 h) programs** (OSD-498 Embryo +1.87, 12 h +2.12) — a rapid DNA-damage/ROS surge resembling the early-germination state, **converging with the µg and GCR-80 "12 h attractor."** - Responses **resolve/flip by 24 h** (repair phase); **low-dose (cGy) effects are weaker and dose-graded**. - *Caveat:* 100 Gy is a mechanistic DNA-damage dose, **not space-relevant** (spaceflight GCR is cGy); acute-γ contrasts are ROS/DNA-damage references, while the cGy studies (OSD-658, OSD-782) are dose-relevant. OSD-508 vs OSD-510 show study-level heterogeneity (interpret per-study).

4.5 Convergence — the radicle growth-point [D × L × A → H]

Three independent lines point to the **radicle apical meristem / root growth-point**: 1. **NMF** induces radicle-apical-meristem-expressed genes (z +7.96) [L×A]; 2. **Microgravity** shows light-gated modulation of the radicle meristem program [D] (cl14: µg-root-dark +1.7 vs µg-root-light –1.6); 3. **High static field** (Jin 2019, 600 mT) regulates radicle growth via auxin/PIN3-AUX1 [L]. The radicle is the first organ to resume growth at germination → a coherent, testable nexus.

4.9 DeepSpace seed-susceptibility atlas (headline) [D×A×L]

(Figure: `results/figures/deepspace_seed_atlas.png`.)

The full perturbation model (21 contrasts spanning **gravity, tropism, low-oxygen, radiation, and magnetic/NMF** families) projected onto germinating-seed cell types, scored as **how many of 5 stressor families significantly target each cell type** (family-level convergence, so radiation's many contrasts get one vote).

Result — the radicle/root tip is the multi-stressor convergence hotspot: - **Radicle apical meristem — 4 / 5 families** (gravity, tropism, radiation, magnetic/NMF; strongest NMF localization z +7.96). - Columella/root-cap (+QC) — 4/5; radicle epidermis — 3/5. All three root-tip cell types rank at the top. - Cotyledon mesophyll — 4/5 (the main non-root hotspot: photosynthetic programs). - (Unassigned clusters cl8/cl11 score high but lack identity → the radicle apical meristem is the top *biologically-defined* hotspot.)

This is the paper's central atlas: it answers *which seed cell types are susceptible to deep-space stressors* — the **root tip is hit by the most independent stressor families**, converging with the NMF localization, light-gated µg, gravitropism, and the hypophysis→radicle-meristem developmental lineage. (`results/deepspace_atlas_results.md`)

Tissue & stage resolution (`results/figures/deepspace_atlas_tissue_stage.png`): at the germination- **stage** level, **12 hsl (early germination) is the most multi-stressor-susceptible window** (4/5 families) — the dry/0 h pole is under-sampled. At the broad developing-**tissue** level the signal is diffuse and maternal-tissue-weighted (Ovule top but only 172 cells), confirming that **cell-type × stage is the informative resolution**. Combined: *radicle/root-tip cell types at early germination* are the hotspot.

5. Integrated model

Space-relevant stressors, read through the seed decoder, do **not** act diffusely — they re-tune specific germinating-seed programs: - a **shared early-germination (12–24 h) attractor** engaged by most stressors; - a **radiation→embryo-provascular** axis (GCR-80); - a **magnetic→radicle-growth-point** axis (NMF localization + µg light-gating + SMF auxin); - a **light gate** on the microgravity response in seed-tissue programs. The developmental bridge shows these predictions are only meaningfully transferable along the **embryo lineage** — the part of the seed that actually carries forward into germination.

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6. Data landscape & limitations

- **Genome-wide NMF is not public.** Two Maffei NNMF microarrays exist — 2022 NNMF time-course (PMC9775259) and 2021 **dose-response** (PMC8080623; 240/40 nT near-null → 60 µT) — but **both state "supplementary tables + data on request"; neither is deposited** in GEO/ArrayExpress. NMF here therefore uses the partial 194-gene panel via localization, **not** the full-transcriptome GSEA used for µg/GCR. *(Author request drafted for both arrays; the NMF arm upgrades to full decoder+bridge columns on arrival.)*
- **Magnetic comparators (high-field) are public**: GSE29787 (Paul/Ferl), PRJNA529956 (Jin SMF 600 mT) — usable as labeled high-field contrasts, not null-field.
- **Dry/0 h pole under-sampled** — germination single-cell starts at 12 h.
- **"Unassigned" germination clusters (8, 11)** lack identity (per the paper) — bridge hits there are soft.
- **Bridge scale artifact** — mixed-scale NES vs ssGSEA; argmax assignments are the robust readout (within-source scaling queued).
- **Concordance, not causation** throughout.

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7. Testable hypotheses (with confidence + falsification)

#	Hypothesis	Tier	Conf.	Falsifying experiment
H1	NMF induces radicle-apical-meristem genes; NNMF germination accelerates radicle emergence	L×A→H	**High**	Germinate Col-0 under NNMF (triaxial Helmholtz); qPCR/in-situ radicle-meristem markers (WOX5, PLT1/2, RGF) + emergence kinetics. Falsify if no meristem-marker induction. `cry1cry2` should abolish.
H2	Microgravity suppresses radicle/seed proliferative programs in a **light-gated** manner	D×A→H	**High**	Spaceflight/clinostat germination, light vs dark; qPCR g2/m + radicle-meristem markers. Falsify if no light×µg interaction.
H3	GCR-80 cGy induces embryo provascular/hypophysis programs	D×A→H	Medium	Dose-series irradiated seed germination; provascular markers (ATHB8, SMXL5, TMO5). Falsify if not dose-dependent.
H4	Only the embryo lineage bridges development→germination; maternal seed coat/endosperm are terminal	A×L→H	Medium	Marker-persistence / lineage tracing across desiccation→imbibition. Falsify if maternal-tissue markers re-express in the germinating embryo.

H5	An early-germination (12–24 h) state is a common transcriptional attractor across µg/GCR(/NMF)	D→H	Medium	Cross-stressor signature-similarity test vs a tissue-matched stress panel. Falsify if stressors diverge in seed-program space.
H6	Null and high magnetic fields converge on the radicle growth-point program	L×D→H	Medium	Overlap NNMF (Maffei) vs SMF (Jin/PRJNA529956) DEGs within radicle-meristem markers. Falsify if no shared core.

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8. Next steps

1. **Genome-wide NMF** (author request: 2021 dose-response + 2022 arrays) → upgrade NMF to full GSEA decoder + a true bridge column; add the dose-response as a magnetic gradient.
2. **Bridge refinement** — within-source scaling; restrict the dev side to the embryo lineage for a cleaner dev→germination map.
3. **Anchor the dry/0 h pole** (germination bulk controls / supplement).
4. **Experimental validation** of H1/H2 (highest confidence).

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9. Appendix — accessions & file manifest

Accessions. Gehring dev = GEO **GSE295007**; germination = ArrayExpress **E-MTAB-12532** / GEO mirror **GSE182331**; stressors = NASA OSDR **OSD-120/678/658** (GLDS-120/612/603); NMF = Maffei **PMC9775259** (2022) + **PMC8080623** (2021), undeposited; high-field comparators = GEO **GSE29787**, SRA **PRJNA529956**.

Key outputs. `panels/panel_library.csv` (+`_annotated`), `panels/germination_cluster_annotations.csv`; `results/tables/decoder_{nes,fdr}_matrix.csv`, `decoder_long.csv`, `bridge_{latent,assignments}.csv`, `nmf_{gene_directions,localization,panel_overlap}.csv`; figures in `results/figures/`; component write-ups `results/{decoder,bridge,nmf}_results_v1.md`. Scripts `scripts/01–08`. Living plan: `README.md`.

Primary references. Gehring et al. *Nat Plants* 2026 (10.1038/s41477-026-02295-8); Liew/Lewsey et al. *Nat Plants* 2024 (10.1038/s41477-024-01771-3); Parmagnani/Maffei *Biomolecules* 2022 (10.3390/biom12121824); Agliassa/Maffei *Sci Rep* 2021 (10.1038/s41598-021-88695-6); Jin et al. *Sci Rep* 2019 (10.1038/s41598-019-50970-y); NASA GeneLab OSD-120/678/658.

Figure plates

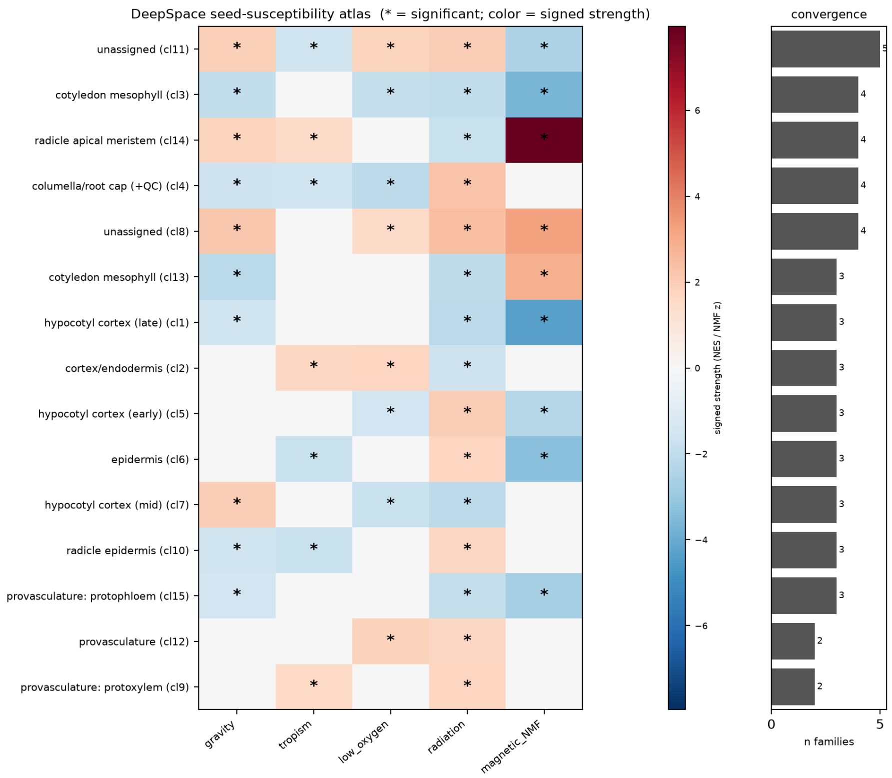


Fig 5 (HEADLINE). DeepSpace seed-susceptibility atlas: germinating-seed cell type x stressor family + multi-stressor convergence (radicle tip = top hotspot).

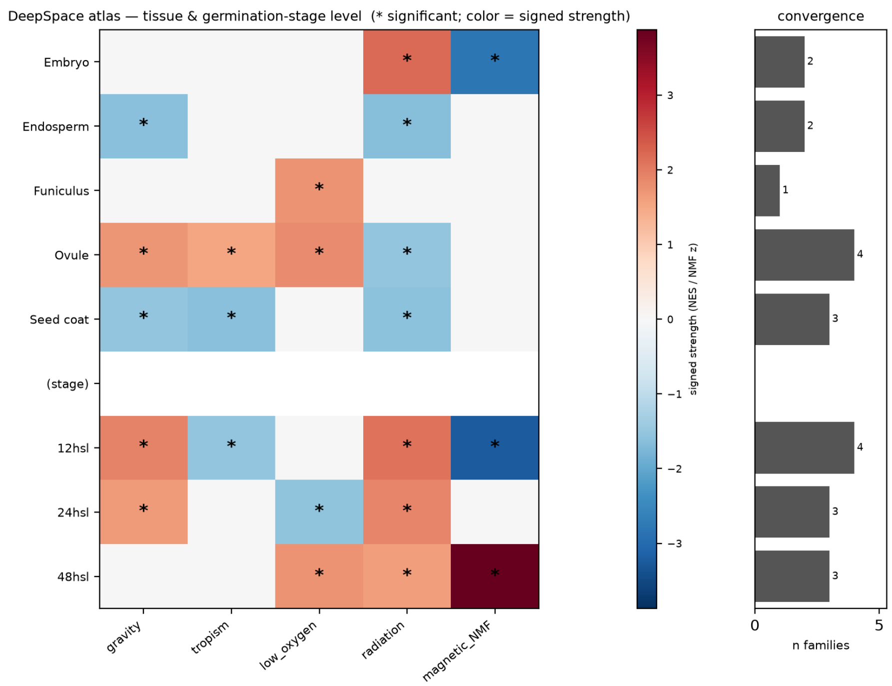


Fig 5b. DeepSpace atlas at tissue + germination-stage level (12h = most multi-stressor stage).

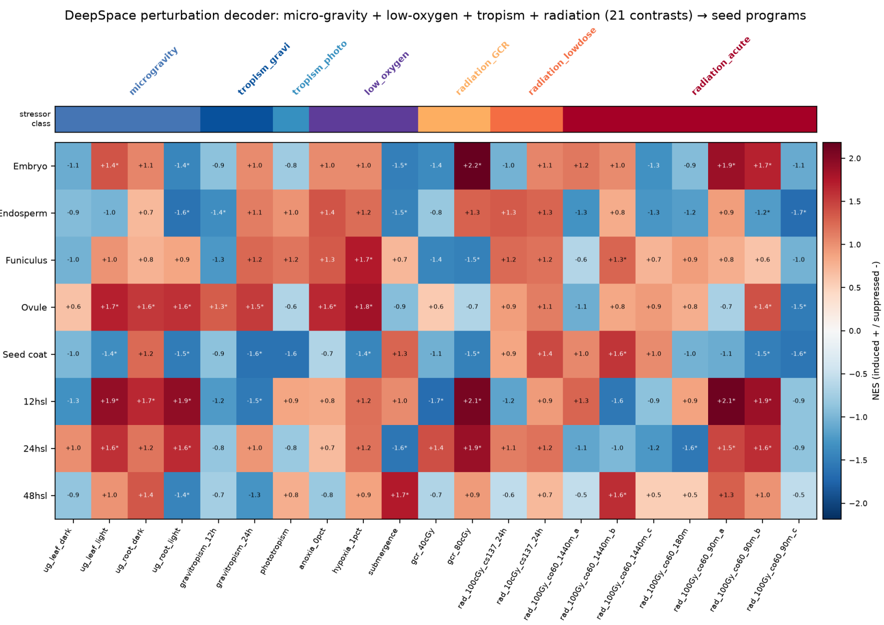


Fig 0. Combined DeepSpace perturbation model: micro-gravity + low-oxygen + tropism + radiation (21 contrasts) -> seed programs.

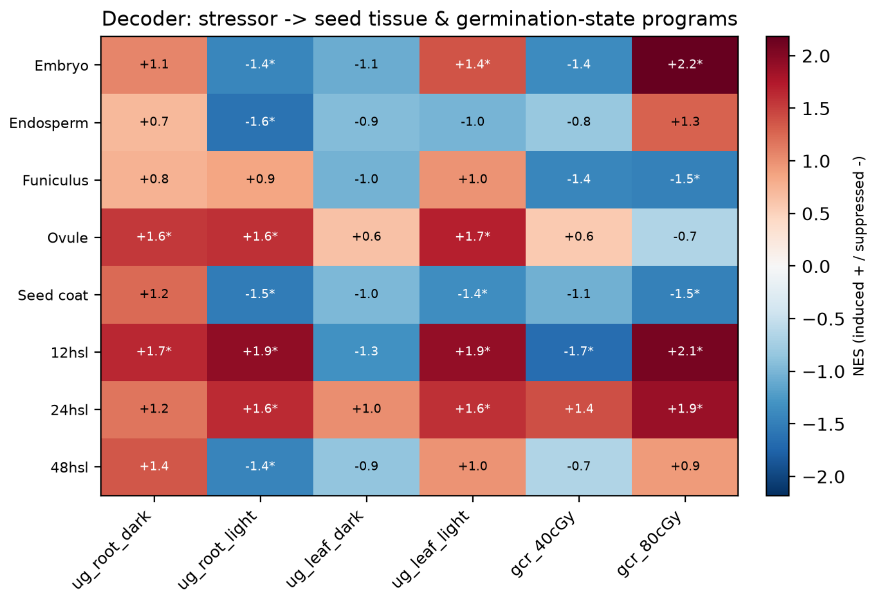
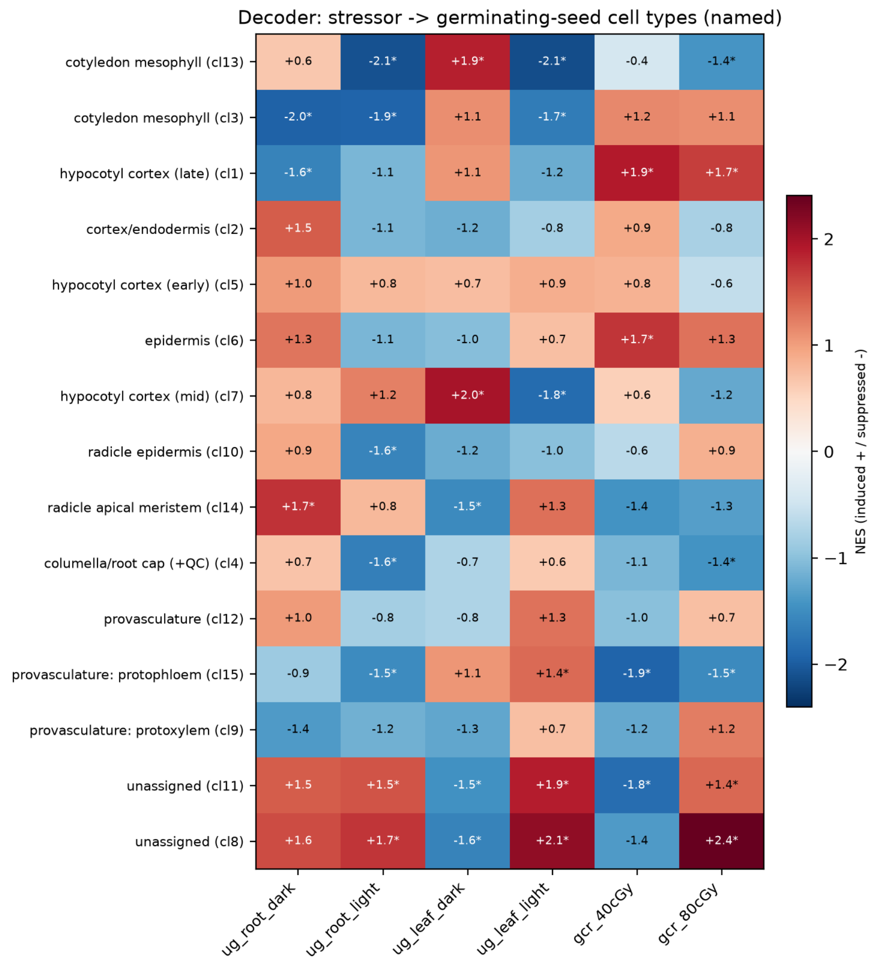


Fig 1. Decoder: stressor -> seed tissue & germination-state programs (NES).



	ug_root_dark	ug_root_light	ug_leaf_dark	ug_leaf_light	gcr_40cGy	gcr_80cGy
cotyledon mesophyll (cl13)	+0.6	-2.1*	+1.9*	-2.1*	-0.4	-1.4*
cotyledon mesophyll (cl3)	-2.0*	-1.9*	+1.1	-1.7*	+1.2	+1.1
hypocotyl cortex (late) (cl1)	-1.6*	-1.1	+1.1	-1.2	+1.9*	+1.7*
cortex/endodermis (cl2)	+1.5	-1.1	-1.2	-0.8	+0.9	-0.8
hypocotyl cortex (early) (cl5)	+1.0	+0.8	+0.7	+0.9	+0.8	-0.6
epidermis (cl6)	+1.3	-1.1	-1.0	+0.7	+1.7*	+1.3
hypocotyl cortex (mid) (cl7)	+0.8	+1.2	+2.0*	-1.8*	+0.6	-1.2
radicle epidermis (cl10)	+0.9	-1.6*	-1.2	-1.0	-0.6	+0.9
radicle apical meristem (cl14)	+1.7*	+0.8	-1.5*	+1.3	-1.4	-1.3
columella/root cap (+QC) (cl4)	+0.7	-1.6*	-0.7	+0.6	-1.1	-1.4*
provasculature (cl12)	+1.0	-0.8	-0.8	+1.3	-1.0	+0.7
provasculature: protophloem (cl15)	-0.9	-1.5*	+1.1	+1.4*	-1.9*	-1.5*
provasculature: protoxylem (cl9)	-1.4	-1.2	-1.3	+0.7	-1.2	+1.2
unassigned (cl11)	+1.5	+1.5*	-1.5*	+1.9*	-1.8*	+1.4*
unassigned (cl8)	+1.6	+1.7*	-1.6*	+2.1*	-1.4	+2.4*

Fig 2. Decoder: stressor -> named germinating-seed cell types.

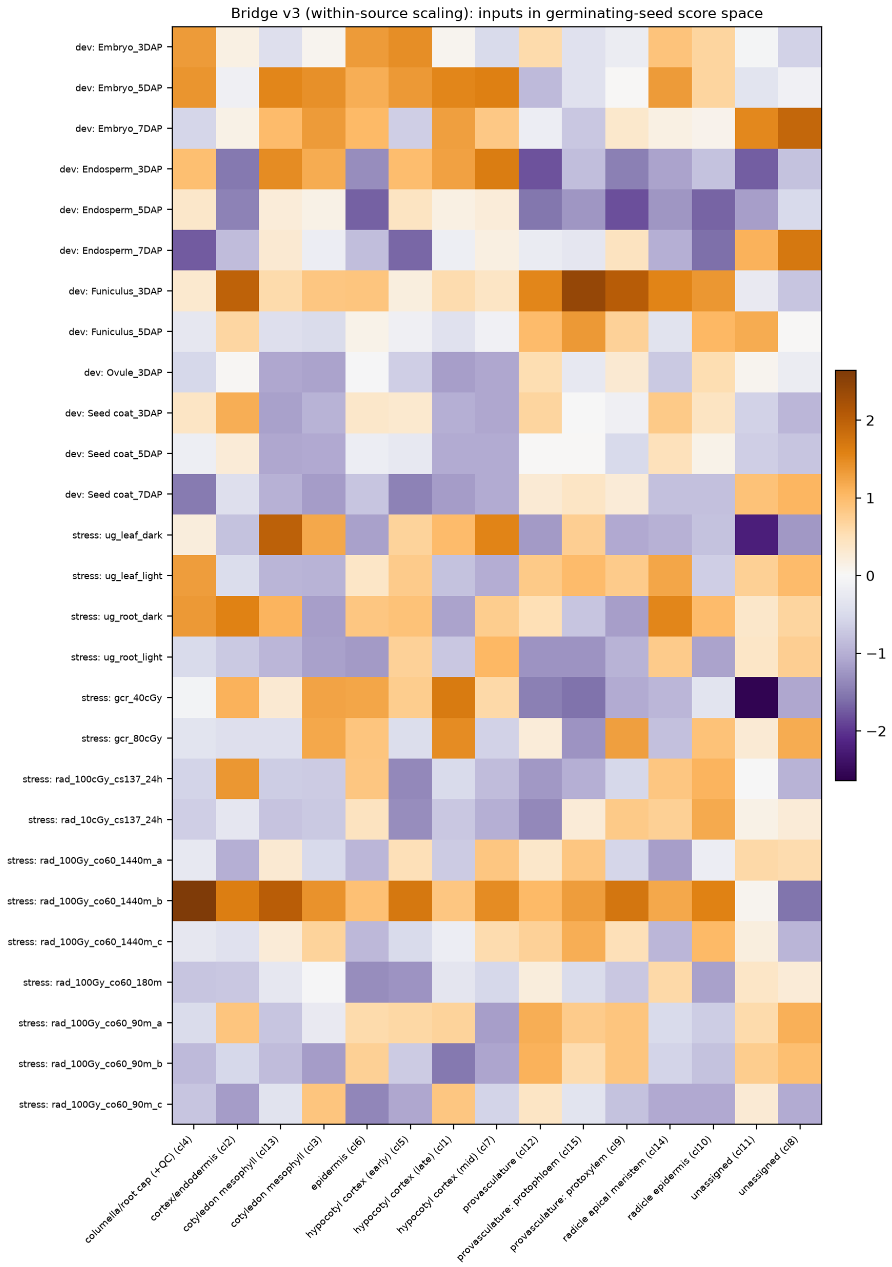


Fig 3. Shared latent v3 (within-source scaled): late-seed-dev + micro-gravity + radiation in germinating-seed score space.

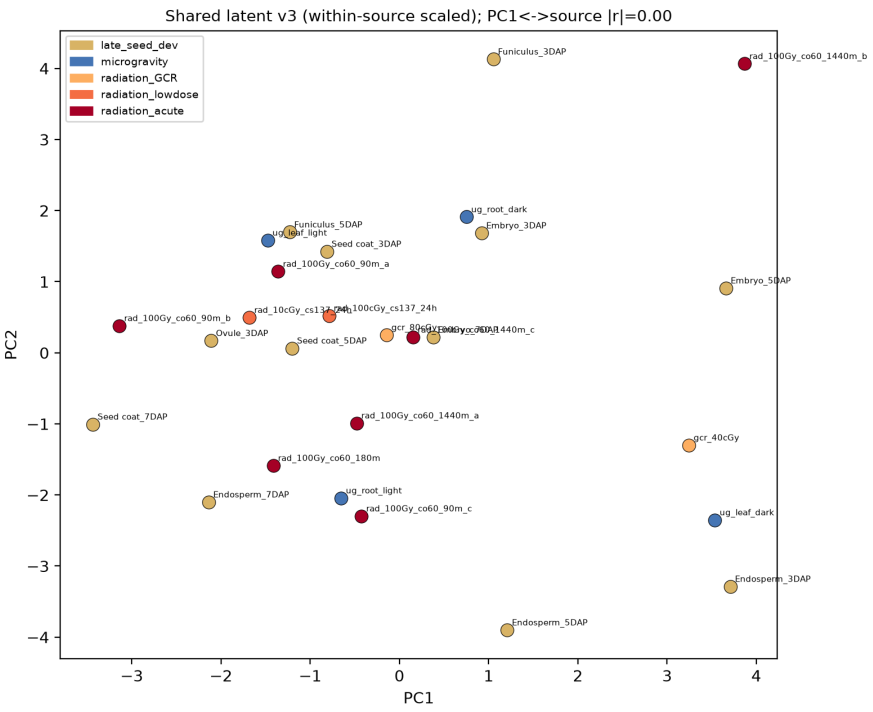


Fig 3b. Shared latent v3 embedding (PCA); PC1<->source |r|=0.00 (artifact removed).

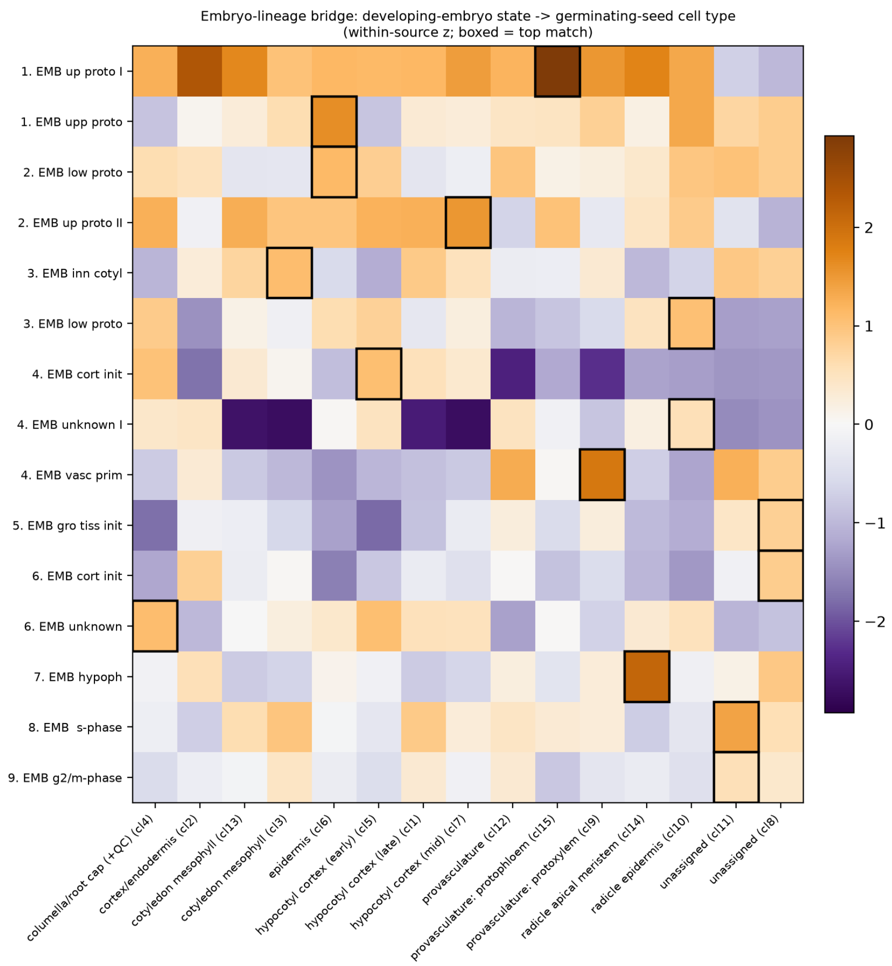


Fig 3c. Embryo-lineage bridge: developing-embryo state -> germinating-seed cell type (boxed = top match).

Fig 4. NMF-responsive gene localization across germinating-seed cell types.